

Sharing & Unit Conversions — Practice

Section A is interactive. Sections B and C are written practice — check them against the separate answer key.

Objective (1 mark each)

Q1 · MCQ**[1]**

To share a quantity x in the ratio $m : n$, the first part is:

- (a) $m \times x$ (b) $m/(m + n) \times x$ (c) x/m (d) $(m + n)/m \times x$

Q2 · MCQ**[1]**

₹100 is shared in the ratio 3 : 2. The larger share is:

- (a) ₹40 (b) ₹50 (c) ₹60 (d) ₹66

Q3 · FILL IN THE BLANK**[1]**

1 litre = _____ mL.

Q4 · MCQ**[1]**

25 °C in Fahrenheit is:

- (a) 45 °F (b) 77 °F (c) 57 °F (d) 90 °F

Q5 · TRUE / FALSE**[1]**

1 hectare = 10,000 square metres.

True False

Q6 · ASSERTION-REASON

[1]

Assertion (A): Dividing 40 kg in the ratio 3 : 1 gives 30 kg and 10 kg.

Reason (R): Each share equals (its ratio number) / (sum of ratio numbers) × total.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Short answer (2 marks each)

Q7

[2]

Divide 42 in the ratio 4 : 3.

Q8

[2]

Convert 25 °C to Fahrenheit, showing the formula you use.

Q9

[2]

The masses of equal volumes of gold and water are in the ratio 37 : 2. If 1 litre of water is 1 kg, what is the mass of 1 litre of gold?

Long answer (3 marks each)

Q10

[3]

A 40 kg mortar mix has sand : cement = 3 : 1. Find the weight of the sand and the cement.

Q11

[3]

Divide ₹360 between two people in the ratio 5 : 4.

Q12

[3]

A plot is 200 ft by 500 ft. Good practice is to apply 10 tonnes of manure per acre (1 acre = 43,560 ft²). How much manure is needed, to the nearest tonne?

Total: 21 marks · Check your work against the answer key.